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National University of Computer and Emerging Sciences
Lahore Campus

Engineering Economics (MG2002)

Date: November 1st 2025

Course Instructor(s)

Prof. Dr. Aftab Ahmad

Sessional-II Exam

Total Time: 1 Hour
Total Marks: 40
Total Questions: 04

Roll No ______________________ Section 3B Student Signature ____________

Do not write below this line

Notes:

i. **Question 1, and Q4c** are to be answered on the Question Paper and the rest on the answer sheet.
ii. Complete Question Paper and **Rough work sheets** are to be returned back along with the answer sheet.
iii. Necessary Compound Interest Factor Tables are available at the last page of the question paper.

CLO # 2: Interpret the concept of time value of money

Q1: Fill in the blanks/select the best choice. **[12]**

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I. What is true about SPCAF?

a. It is a cash inflow
b. ✓ It is a future amount
c. It is a cash outflow
d. None of the above

II. What is not true about arithmetic gradient?

a. It is the amount of change.
a. It is an increase by a constant amount
b. ✓ It is a percentage decrease
c. None of the above

III. The interest rate for 4 years is 10%, and for 6 years is 14%. What is the interest rate for 5 years using linear interpolation?

a. 11%
b. ✓ 12%
c. 13%
d. 15%

IV. If the growth rate *g* equals the interest rate *I*, the present worth of a geometric gradient series becomes:

a. Infinite
b. Zero
c. ✓ None of the above

V. Upon the investment of Rs. 10 million, an engineering firm expects annual return at 8% while 14% on Rs. 18 million. Estimate the expected return if the firm invests Rs. 15 million, using linear interpolation.?

a. 9%
b. 10%
c. 11%
d. 12%

VI. The F/G factor values can be derived by multiplying:

a. (P/F) and (A/G) factor values
b. (F/P) and (A/G) factor values
c. (P/F) and (P/G) factor values
d. (F/P) and (P/G) factor values

VII. If a uniform series begins immediately (first payment at time 0), it is called a:

a. Shifted series
b. Deferred series
c. Perpetual series
d. Annuity due

VIII. The first payment of a geometric gradient series is Rs. 10,000. The payments increase by 8 % per year for 4 years. The interest rate is 10 %. Find the present worth (P) of the series.

a. Rs 32,000
b. Rs 33,000
c. Rs 34,000
d. Rs 35,000

IX. A payment of Rs 5,000 occurs every year for 4 years, starting at the end of Year 3. Interest rate i = 10%. Find the present worth (P).

a. Rs 13,000
b. Rs 14,500
c. Rs 15,000
d. Rs. 16,000

X. USPWF is also referred to as sinking fund factor. (True / False)

XI. Sinking funds and capital recovery are both financial concepts. (True / False)

XII. A shifted series beginning at the end of year 3 can be treated as a regular series that starts at year 1 and is discounted two years back to the present. (True / False)

CLO # 2: Interpret the concept of time value of money

Q2:

(a) Determine the difference in the present worth values of the following two commodity contracts at an interest rate of 8% per year. Contract 1 has a cost of $20,000 in year 1; costs will escalate at a rate of 4% per year for 10 years. Contract 2 has the same cost in year 1, but costs will escalate at 6% per year for 11 years. Step wise complete working is to be shown in your answer. **[5]**

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(b) **If Ali made a $50,000 investment in a friend's business and received $80,000 5 years later. Using formula and factor value table, determine the rate of return. Step wise complete working is to be shown in your answer.** **[5]**

(c) **Determine the P/A factor value for *i* = 7.75% and n = 5 years, using formula as well as table.** **[3]**

CLO # 3: Analyze cost effectiveness of alternatives using engineering economy methods

Q3: Define the following terms: **[5]**

(a) **Discounted Cash flows**
(b) **Mutually Exclusive Projects**
(c) **Independent projects:**
(d) **DN Alternative or Proposal**

CLO # 1: Recognize the fundamental economic and financial considerations involved in engineering projects

Q4:

(a) **Briefly introduce "Professional or engineering ethics".** **[3]**
(b) **List the considerations in choosing an organizational type (form of business).** **[2]**
(c) **Fill the empty 6 cells of the table:** **[5]**

Aspect	Statutory Corporation	Government-Owned Company
1. Legal Basis	In this we have to approve our corporation on legal level as well. Some owend by group of people who work privatly. They decide about the future of cooporation. Also some government members are included which took part in decision making.	Company owned and run by government. Government is going to decide about the future of company. Public also work there but could not take part in decision making process.
2. Ownership	Owned by group of people it also contain some people from government. Like a mixup of private and government members.	Completely owned by government only. People work there as servant but not able to took part in decision making.
3. Example of Authority	PC Hotel	PTCL PTA PCB